

Research Interests

My current research focuses on aligning generative models through principled reward engineering. I study the fundamental limits of reward model uncertainty and investigate how these limits affect post-training alignment protocols such as RLHF and inference-time alignment. I also study group fairness in LLM alignment, with the goal of ensuring equitable treatment of diverse user groups in human feedback pipelines.

Education

2019–2024 **Rensselaer Polytechnic Institute.**

- Ph.D. candidate, department of Electrical Computer & Systems Engineering
- GPA: 3.95/4
- Advisor: Ali Tajer

2017–2019 : **Indian Institute of Technology, Kharagpur.**

- M.Tech., Department of Electronics and Electrical Communication Engineering
- GPA: 9.19/10
- Advisor: Mrityunjay Chakraborty

2013–2017 : **West Bengal University of Technology.**

- B.Tech., Department of Electronics and Communication Engineering
- GPA: 9.05/10
- Advisor: Krishanu Datta

Work Experience

12.24 – **Imperial College London, London.**

- present
- Informed-AI postdoctoral research associate
 - Advisor: Deniz Gündüz

06.24 – 08.24 **IBM Research, NY.**

- Research Internship
- Project: Bandit-based Exploration for Prompt Engineering in LLMs
- Collaborators: Djallel Bouneffouf, Miao Liu
- Manager: Prasanna Sattigeri

06.23 – 08.23 **IBM Research, NY.**

- Research Internship
- Project: Group testing for combinatorial bandits
- Collaborators: Shashanka Ubaru, Keerthiram Murugesan and Karthikeyan Shanmugam
- Manager: Lior Horesh

06.21 – 08.21 **IBM Research, NY.**

- Research Internship
- Project: Data-aware client selection in federated learning
- Collaborators: Theodoros Salonidis, Shiqiang Wang and Georgios Kollias
- Manager: Theodoros Salonidis

Publications

Preprints

- 2026 **Preference-centric Bandits: Optimality of Mixtures and Regret-efficient Algorithms**, *IEEE Transaction on Information Theory* (under revision, submitted 2025).
M. Tatli, **A. Mukherjee**, Prashanth LA, K. Shanmugam and A. Tajer

Conferences (* denotes equal contribution)

- ICLR 2026 **Test-time Verification via Optimal Transport: Coverage, ROC, & Sub-optimality**, *Proc. International Conference on Learning Representations*, Rio de Janeiro, Brazil, April 2026.
A. Mukherjee, M. Bullo, D. Basu and D. Gündüz
- AAAI 2026 **SharedRep-RLHF: A Shared Representation Approach to RLHF with Diverse Preferences**, *Proc. Association for the Advancement of Artificial Intelligence (Oral)*, preliminary version presented at the Foundations of Post-training workshop, COLT 2025), Singapore, January 2026.
A. Mukherjee*, M. Bullo* and D. Gündüz
- AISTATS 2025 **Risk-sensitive Bandits: Arm Mixture Optimality and Regret-efficient Algorithms**, *Proc. International Conference on Artificial Intelligence and Statistics*, May 2025.
M. Tatli*, **A. Mukherjee***, Prashanth LA, K. Shanmugam and A. Tajer
- ISIT 2024 **BAI in Exponential Family: Efficiency and Optimality**, *Proc. International Symposium on Information Theory*, July 2024.
A. Mukherjee and A. Tajer
- ISIT 2024 **Improved Bound for Robust Causal Bandits with Linear Models**, *Proc. International Symposium on Information Theory*, accepted for publication.
Z. Yan, **A. Mukherjee**, B. Varici and A. Tajer
- ISIT 2022 **SPRT-based Best Arm Identification in Stochastic Bandits**, *Proc. International Symposium on Information Theory*, Helsinki, Finland, June 2022.
A. Mukherjee and A. Tajer
- NeurIPS 2021 **Mean-based Best Arm Identification in Stochastic Bandits under Reward Contamination**, *Proc. Advances in Neural Information Processing Systems*, virtual, December 2021.
A. Mukherjee, A. Tajer, P. Das and P.-Y. Chen
- ISIT 2021 **Active Binary Classification of Random Fields**, *Proc. International Symposium on Information Theory*, Melbourne, Australia, July 2021.
A. Mukherjee, A. Tajer, P. Das and P.-Y. Chen
- ICASSP 2021 **Active Estimation from Multimodal Data**, *Proc. International Conference on Acoustics, Speech and Signal Processing*, Toronto, Canada, June 2021.
A. Mukherjee, A. Tajer, P. Das and P.-Y. Chen

Journals

- TMLR 2025 **Combinatorial Multi-armed Bandits: Arm Selection via Group Testing**, *Transactions on Machine Learning Research* (accepted for publication), May 2025.
A. Mukherjee, S. Ubaru, K. Murugesan, K. Shanmugam and A. Tajer
- T-IT 2024 **Best Arm Identification in Stochastic Bandits: Beyond β -optimality**, *IEEE Transaction on Information Theory* (accepted for publication), November 2024.
A. Mukherjee and A. Tajer
- T-IT 2024 **Optimal Best Arm Identification with Fixed Confidence in Restless Bandits**, *IEEE Transaction on Information Theory* (accepted for publication), June 2024.
P. N. Karthik, V. Y. F. Tan, **A. Mukherjee** and A. Tajer

- JSAIT 2024 **Robust Causal Bandits for Linear Time-varying Models**, *IEEE Journal on Selected Areas in Information Theory* (accepted for publication), February 2024.
Z. Yan, **A. Mukherjee**, B. Varici and A. Tajer
- JSAIT 2023 **SPRT-based Efficient Best Arm Identification in Stochastic Bandits**, *IEEE Journal on Selected Areas in Information Theory*, vol. 4, pp. 128-143, July 2023.
A. Mukherjee and A. Tajer
- TSP 2022 **Active Sampling of Multiple Sources for Sequential Estimation**, *IEEE Transactions on Signal Processing*, vol. 70, pp.4571-4585, July 2022.
A. Mukherjee and A. Tajer
- TSP 2020 **ImdLMS: An Imputation based LMS algorithm for Linear System Identification with Missing Input Data**, *IEEE Transactions on Signal Processing*, vol. 68, pp. 2370-2385, 2020.
S. Mukhopadhyay and **A. Mukherjee**

Teaching Experiences

- Spring 2024 **Teaching Assistant**, *Electronic Instrumentation*, (ENGR 2300), RPI.
Fall 2023 **Teaching Assistant**, *Electronic Instrumentation*, (ENGR 2300), RPI.
Fall 2022 **Teaching Assistant**, *Electronic Instrumentation*, (ENGR 2300), RPI.

Fellowships & Awards

- 2025 **Allen B. Dumont** (RPI ECSE doctoral dissertation award)
2022 **Winner** of the ISIT Information Theoretic Duets (along with Rajarshi Saha (currently at Amazon))
2019 – 2020 Recipient of the **B. Jayant Baliga '74 Graduate Student Fellowship Award** at RPI
2017 Recipient of the **MHRD PG Fellowship** through GATE

Invited Talks

- May 2025 **RLHF with Diverse Preferences**, *Informed-AI Summer School*, (University of Bristol).
October 2024 **Risk-sensitive Bandits**, *IIT Bombay*, (CMinDs).
October 2024 **Risk-sensitive Bandits**, *IISc Bengaluru*, (CNI).
March 2024 **Stochastic Bandits: Complexity and Optimality**, *Imperial College London*, (Information Processing and Communications Lab).
March 2023 **Security and Safety in Multi-Armed Bandits**, *Vector Institute*, (University of Toronto).
Spring 2022 **Contaminated Best Arm Identification in Stochastic Bandits**, *Conference on Information Sciences and Systems (CISS) 2022*, (Princeton).

Technical Skills

- Languages Python, MATLAB
Libraries PyTorch, NumPy, Pandas
LLM Tooling HuggingFace Transformers, TRL, Weights & Biases (wandb)

Academic Services

- 2025 Reviewer for ICLR 2026
2025 Reviewer for AAAI 2026
2025 Reviewer for ICML 2025
2024 Reviewer for NeurIPS 2024
2024 Reviewer for IEEE ISIT 2024

2024 Reviewer for IEEE Journal on Selected Areas in Information Theory
2023 Reviewer for AISTATS 2024
2023–2024 Reviewer for IEEE Transactions on Information Theory
2023 – 2024 Reviewer for IEEE Transactions on Mobile Computing
2023 Reviewer for IEEE ISIT 2023
2022 – 2023 Reviewer for IEEE Transactions on Communication
2022 Reviewer for AISTATS 2023
2021 Reviewer for AISTATS 2022
2021 Reviewer for AAAI 2021
2021 Reviewer for NeurIPS 2021
2019 – 2022 Reviewer for IEEE Transactions on Signal Processing

Mentoring

Marcello Bullo, fourth year Ph.D. student, Imperial College London.

Meltem Tatlı, second year Ph.D. student, RPI.

Zirui Yan, fourth year Ph.D. student, RPI.

Work Authorization

Authorized to work in the United Kingdom. Currently on a Global Talent Visa (endorsed for academic research).

Referees

Prof. Ali Tajer

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Academic Affairs, Department of
Electrical, Computer & Systems Engineering*
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